FOURTH SEMESTER M.Sc. DEGREE EXAMINATION, JUNE 2018

(CUCSS)

Mathematics

MT 4C 15—FUNCTIONAL ANALYSIS—II

(2010 Admissions)

Time : Three Hours

Maximum : 36 Weightage

Part A

Short Answer Questions (1-14).
Answer all *questions.*
Each question carries 1 *weightage.*

1. Let X, Y, Z be normed spaces, $F : X \to Y$ be a continuous linear map and $G : Y \to Z$ be a closed linear map. Prove that G o F is a closed linear map.

2. Let X be a normed space and let P be a A projection on X. If the range space $\mathcal{R}(P)$ and the zero space $\mathcal{Z}(P)$ are closed subspaces of X, then prove that P is a closed map.

3. Let A be a bounded operator on a normed space X. Prove that :

$$\sigma_e(A) \subset \sigma_e(A) \subset \sigma(A).$$

4. Let X be a normed space over a field K, let $A, B \in BL(X)$ and let $0 \neq k \in K$. Prove that $k \in \sigma(AB)$ if and only if $k \in \sigma(BA)$.

5. Let X, Y be normed spaces and let $F \in BL(X, Y)$. Prove that :

$$\|F'\| = \|F\| = \|F''\|.$$

6. Prove that the dual of a reflexive normed space is reflexive.

7. Let E be a subset of the Hilbert space H. Prove that $E^{\perp\perp}$ is the closure of span of E.

8. Find the orthogonal projection on the Hilbert space $\mathbb{R}^2$ onto the closed subspace :

$$\{(x, y) \in \mathbb{R}^2 : x + y = 0\}.$$

9. Define weak convergence. Prove that convergence implies weak convergence.

10. Let H be a Hilbert space and let $A \in BL(H)$. Prove that

$$\|A^*A\| = \|A^*\|^2 = \|AA^*\|.$$

11. Let H be a Hilbert space. Prove that the set of all unitary operators in BL (H) is a group.

12. Is composition of positive operators on a Hilbert space positive ? Justify your answer.

13. Let A be a bounded operator on a Hilbert space H. Prove that $\sigma_e(A) \subset \omega(A)$, where $\omega(A)$ is the numerical range of A.

14. Let A be a bounded self adjoint operator on a non-trivial Hilbert space H. Prove that the scalar

$$\sup_{x \in H, \|x\|=1} \langle A(x), x \rangle$$

is an approximate eigen value of A.

(14 × 1 = 14 weightage)

Part B

Answer any **seven** *from the following ten questions* (15-24).
Each question has weightage 2.

15. Let X and Y be normed spaces and let $F : X \to Y$ be a linear map. Prove that F is an open map if and only if there exists some $\gamma > 0$ such that for every $y \in Y$, there is some $x \in X$, with $F(x) = y$ and $\|x\| \leq \gamma \|y\|$.

16. Let X be a Banach space, $A \in BL(X)$ and let $\|A^p\| < 1$ for some positive integer p. Prove that the bounded operator I – A is invertible.

17. Let X be a Banach space over **K** and $A \in BL(X)$. Prove that $\sigma(A)$ is a compact subset of **K**.

18. Let X be an inner product space and let $f \in X'$. If $\{u_1, u_2, ...\}$ is an orthonormal set in X, then prove that :

$$\sum_n |f(u_n)|^2 \leq \|f\|^2.$$

19. Let H be a Hilbert space. Prove that there exists a surjective linear isometry from H' into H.

20. Let y be a fixed element in an inner product space X over a field K. Define $f : X \to K$ by $f(x) = \langle x, y \rangle$. Prove that $f \in X'$ and $\|f\| = \|y\|$.

21. Prove that the set of all unitary operators on a Hilbert space H is a closed subset of BL (H).

22. Let H be a non-trivial Hilbert space over a field K. If $A \in BL(H)$ is self adjoint, then prove that $\|A\| = \sup \{|k| : k \in \sigma(A)\}$.

23. Let A be a bounded operator on a Hilbert space H. If A is compact, then prove that A^* is compact.

24. Let A be a compact operator on a non-trivial Hilbert space H. Prove that every non-zero approximate eigen value of A is an eigen value of A.

(7 × 2 = 14 weightage)

Part C

Answer any **two** *from the following four questions* (25-28).
Each question has weightage 4.

25. Let X and Y be Banach spaces and let $F : X \to Y$ be a closed linear map. Prove that F is continuous.

26. State and prove projection theorem.

27. Let H be a Hilbert space, G be a subspace of H and let g be a continuous linear functional on G. Prove that there is a continuous linear functional f on H such that $f_{|G} = g$ and $\|f\| = \|g\|$.

28. Let A be a non-zero compact self adjoint operator on a Hilbert space H over a field K. Prove that there exist a finite or infinite sequence (s_n) of non-zero real numbers with $|s_1| \geq |s_2| \geq \ldots$ and an orthonormal set $\{u_1, u_2, \ldots\}$ in H such that :

$$A(x) = \sum_n s_n \langle x, u_n \rangle u_n$$

for all $x \in H$.

(2 × 4 = 8 weightage)